FORMOTEROL + BUDESONIDE

MART: maintenance + reliever

Rinse mouth

CHEMICAL NAME Formoterol fumarate: (RR,SS)-N-[2-hydroxy-5-[1-hydroxy-2-[[2-(4-methoxyphenyl)-1-methylethyl]amino]ethyl]phenyl]formamide + budesonide

CLASS Combination inhaler: fast-onset LABA + inhaled corticosteroid

TRADE NAMES Symbicort · DuoResp Spiromax · Fobumix

THERAPEUTIC USES / INDICATIONS

- Asthma; maintenance, or maintenance and reliever therapy (MART) / as-needed anti-inflammatory reliever (mild asthma, GINA).
- COPD maintenance.

ADMINISTRATION

- **Turbohaler/Spiromax** (dry powder): no shaking; exhale away, inhale fast and deep.
- Rinse mouth after maintenance doses.
- Formoterol's fast onset (1–3 min) allows reliever use in MART.

DOSAGES

- **Maintenance (200/6):** 1–2 inhalations twice daily.
- **MART (200/6):** 1 inhalation twice daily (or 2 once daily) + 1 inhalation as needed; max 8 (occasionally 12) inhalations/day.
- **As-needed only (mild asthma):** 1 inhalation when needed.
- **COPD (400/12):** 1 inhalation twice daily.

PHARMACOKINETICS

- **Formoterol onset:** 1–3 min; duration 12 h.
- Budesonide: high first-pass metabolism.
- Both renally and hepatically cleared.

NURSING MANAGEMENT

- Clarify whether patient is on MART or fixed maintenance.
- Check technique (Turbohaler: twist and click, inhale forcefully).
- Monitor total daily inhalations; >8/day needs review.
- Inspect mouth.

PHARMACODYNAMICS

- Formoterol: rapid, long-acting **β2 agonist**.
- Budesonide: anti-inflammatory glucocorticoid.
- Each reliever dose also delivers steroid → fewer severe exacerbations.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tremor, palpitations, headache.
- Oral thrush, hoarseness.
- Throat irritation.

ADVERSE EFFECTS (SERIOUS)

- Hypokalaemia, tachyarrhythmias (high doses).
- Paradoxical bronchospasm.
- Pneumonia (COPD).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity (lactose).
- **Caution:** cardiovascular disease, thyrotoxicosis, diabetes, TB.

KEY INTERACTIONS

- **β-blockers:** antagonism.
- **Strong CYP3A4 inhibitors:** ↑ budesonide.
- **Other LABAs:** avoid duplication.

PATIENT EDUCATION

- If on MART, use the same inhaler for relief — no separate blue inhaler needed.
- Seek urgent help if you need more than 8 inhalations a day or relief is short-lived.
- Rinse mouth after morning and evening doses.

KEY POINTS

- Fast-onset LABA + ICS.
- Can be used as maintenance and reliever (MART).
- Max 8 (12) inhalations/day.
- Rinse mouth.